

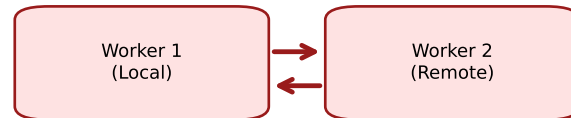
The Distributed GRL Dilemma and the EMO Relational Solution

Overcoming the communication bottleneck and boundary accuracy degradation via Lakehouse-native CaaN abstractions

(a) Synchronized Distributed GRL

- Full-graph state split across nodes
- Heavy inter-worker RPC communication
- Remote neighbor fetching every epoch
- High cluster cost & complex GPU sync

Remote Parameter Servers

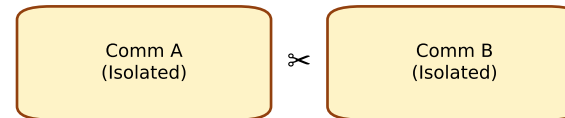


⚠️ Heavy Network RPCs: 28-46 GB

(b) Naive Community Partitioning

- Complete graph partitioning
- 0.0 GB inter-worker network traffic
- BUT severed boundary edges
- Severe Boundary Accuracy Loss (-40%)
- Minor community degeneracy

Severed Cross-Community Edges

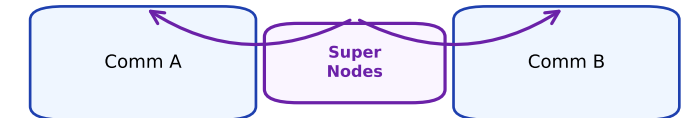


⚠️ Accuracy Drops from 94.5% → 52.9%

(c) EMO: Partition-Centric + CaaN Recovery

- Transactional Delta Lake tables (ACID)
- Decoupled PySpark UDF training
- 0.0 GB inter-worker training traffic
- Relational CaaN super-node broadcast
- Restores 97.8% full-graph accuracy

Super-Node Context Broadcast



✓ 0.0 GB Network | Recovers 92.7% Acc